

Nonlinear Kapitza Problem

Reduction, Pitchfork Bifurcation, S-Curve, and Reproducible Computation

Detailed report in the teacher's parameter convention

Abstract

This report gives a self-contained analysis of the one-dimensional nonlinear Kapitza transmission problem implemented in the accompanying Python and MATLAB programs. The piecewise-affine temperature field reduces the transmission problem to one scalar fixed-point equation. An exponential interface conductance makes that equation exactly equivalent to $x = \tanh(\alpha x/2 + h)$. In the symmetric case $h = 0$, the universal critical value is $\alpha_c = 2$: one solution exists for $\alpha \leq 2$, whereas three solutions exist for $\alpha > 2$, producing a supercritical pitchfork. A nonzero bias h rescales the conductance and unfolds the pitchfork into an S-curve with two analytic folds. Numerical root isolation, fixed-point intersections, cobweb paths, branch classification, and heat-flux intersections are described in direct correspondence with the code.

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1 Transmission problem and physical interval

Let a denote the scalar parameter of the piecewise-affine steady temperature field. The interface jump and steady heat flux satisfy

$$s = [u] = 1 - (1 + \delta)a, \quad q = \delta a = \beta(s)s, \quad \delta > 0, \quad (1)$$

where $\beta(s) > 0$ is the nonlinear Kapitza conductance.

The physical interval is not an additional bifurcation assumption. For the normalized boundary temperatures, the one-dimensional maximum principle keeps the interface traces between the boundary values. In the reduced equations the same conclusion is visible algebraically: positive conductance and positive heat flow give $a > 0$ and $s > 0$; then $s = 1 - (1 + \delta)a$ gives

$$0 < a < \gamma, \quad 0 < s < 1, \quad \gamma := \frac{1}{1 + \delta}. \quad (2)$$

Thus the interval in a follows directly from the maximum-principle interpretation and the interface balance; no separate change-of-variable derivation is needed to justify it.

Solving (1) for a , with $s = 1 - (1 + \delta)a$, yields the scalar map

$$a = F(a) := \frac{\beta(1 - (1 + \delta)a)}{\delta + (1 + \delta)\beta(1 - (1 + \delta)a)}. \quad (3)$$

Every fixed point of F corresponds to one steady transmission state.

2 Teacher's convention and exact inverse construction

The teacher's short Scilab code uses the variable name `d`; throughout this project `delta` denotes the same material parameter δ . The implemented conductance is

$$\beta_h(s) = \frac{\delta}{1 + \delta} \exp[\alpha(1 - 2s) + 2h]. \quad (4)$$

The symmetric law is obtained by setting $h = 0$. It is positive, decreasing, and convex:

$$\partial_s \beta_h = -2\alpha \beta_h < 0, \quad \partial_{ss} \beta_h = 4\alpha^2 \beta_h > 0. \quad (5)$$

Substituting (4) into (3) and using $E = \exp[\alpha(1 - 2s) + 2h]$ gives

$$F_{\alpha,h}(a) = \gamma \frac{E}{1 + E} \quad (6)$$

$$= \frac{\gamma}{2} \left[1 + \tanh\left(\frac{\alpha}{2}(1 - 2s) + h\right) \right]. \quad (7)$$

This identity is the precise link between the nonlinear transmission law and the sigmoid fixed-point map used in both programs.

The relevant lines of the teacher's code are therefore represented exactly by the following project notation:

```
delta = 0.5; % Teacher's variable d
alpha = 1.5;
a = linspace(0, 1/(1+delta), 2000);
beta = @(s) delta/(1+delta).*exp(alpha*(1-2*s));
F = @(a) beta(1-(1+delta).*a) ./ ...
    (delta + (1+delta).*beta(1-(1+delta).*a));
plot(a,F(a),a,a)
```

For $\delta = 0.5$ and $\alpha = 1.5$, the unique intersection is

$$a = \frac{1}{3}, \quad s = \frac{1}{2}, \quad \beta = \frac{1}{3}, \quad q = \frac{1}{6}. \quad (8)$$

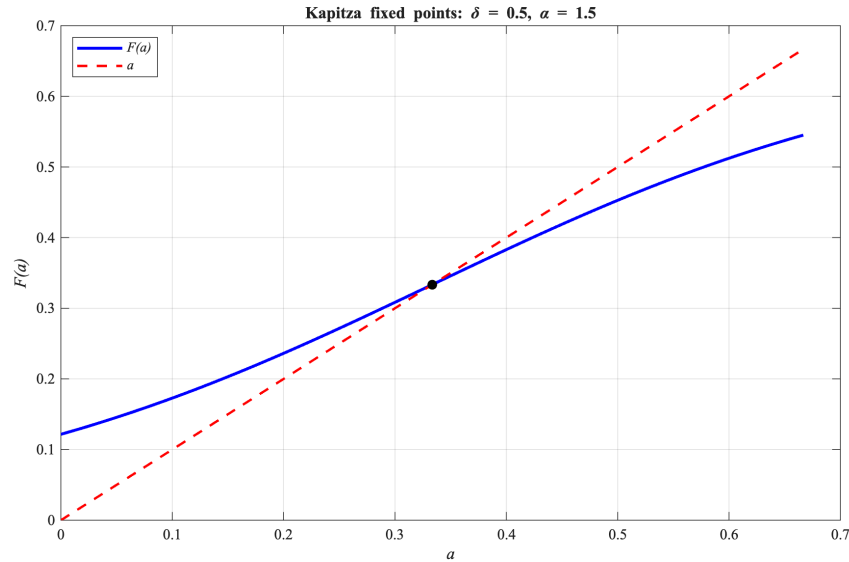


Figure 1: MATLAB reproduction of the teacher's subcritical fixed-point example.

3 Normalized equation and number of solutions

Introduce the normalized state

$$x = \frac{2a}{\gamma} - 1 = 2(1 + \delta)a - 1, \quad -1 < x < 1. \quad (9)$$

Then $s = (1 - x)/2$, so $1 - 2s = x$. Equation (7) becomes

$$x = \tanh\left(\frac{\alpha}{2}x + h\right). \quad (10)$$

The material parameter δ determines the conversion between x and a , but it drops out of the normalized root equation. Consequently the critical value in the teacher's convention is independent of δ .

For $h = 0$, write $k = \alpha/2$ and $G(x) = \tanh(kx) - x$. The function is odd and

$$G'(x) = k \operatorname{sech}^2(kx) - 1. \quad (11)$$

If $k \leq 1$, then G is nonincreasing on $x > 0$, strictly decreasing away from the critical origin, and $x = 0$ is the unique root. If $k > 1$, then $G'(0) > 0$, while $G(1) = \tanh(k) - 1 < 0$; the derivative changes sign once on $x > 0$, hence there is exactly one positive root. Odd symmetry gives one negative root. Therefore

$$\begin{array}{ll} 0 < \alpha \leq 2 : & \text{one steady state,} \\ \alpha > 2 : & \text{three steady states.} \end{array} \quad (12)$$

The central branch is

$$x = 0, \quad a = \frac{1}{2(1 + \delta)}, \quad s = \frac{1}{2}. \quad (13)$$

For $x \neq 0$, the two outer branches have the exact parametrization

$$\alpha(x) = 2 \frac{\operatorname{artanh}(x)}{x}, \quad a(x) = \frac{1 + x}{2(1 + \delta)}, \quad 0 < |x| < 1. \quad (14)$$

Since $\operatorname{artanh}(x)/x = 1 + x^2/3 + O(x^4)$, the outer branches emerge continuously at $\alpha = 2$, which is the geometry of a supercritical pitchfork.

4 Iteration stability and cobweb interpretation

For the scalar iteration $x_{n+1} = \tanh(\alpha x_n/2 + h)$, the multiplier at a fixed point is

$$\lambda = \frac{\alpha}{2} \operatorname{sech}^2\left(\frac{\alpha}{2}x + h\right) = \frac{\alpha}{2}(1 - x^2). \quad (15)$$

A fixed point is attracting when $|\lambda| < 1$ and repelling when $|\lambda| > 1$. In the symmetric problem, the central multiplier is $\lambda = \alpha/2$. Thus the central branch is attracting for $\alpha < 2$ and repelling for $\alpha > 2$. At $\alpha = 2$, linearization is neutral. Nevertheless, $\tanh x = x - x^3/3 + O(x^5)$, so nearby iterates approach the origin only algebraically rather than geometrically.

Along either outer branch, $\alpha/2 = \operatorname{artanh}(x)/x$, and

$$\lambda = (1 - x^2) \frac{\operatorname{artanh}(x)}{x} < 1, \quad 0 < |x| < 1. \quad (16)$$

Hence both outer branches are attracting for this iteration. In every bifurcation figure, solid curves represent attracting branches and dashed curves represent repelling branches.

This stability statement concerns the numerical fixed-point iteration. It is not automatically the physical stability of a time-dependent heat equation; that question would require specifying and linearizing an evolution model.

5 Numerical example at $\delta = 0.5, \alpha = 3$

For the supercritical symmetric example, both codes recover the three states in Table 1. The identity $q = \delta a = \beta(s)s$ holds to numerical precision.

Table 1: Symmetric roots for $\delta = 0.5$ and $\alpha = 3$.

x	a	s	$\beta(s)$	q	λ	Iteration
-0.858560	0.047147	0.929280	0.025367	0.023573	0.394313	attracting
0.000000	0.333333	0.500000	0.333333	0.166667	1.500000	repelling
0.858560	0.619520	0.070720	4.380078	0.309760	0.394313	attracting

The term *central branch* simply means the symmetry-preserving family $x = 0$, equivalently $a = 1/[2(1 + \delta)]$, which exists for every value of α . It is not a separate material law. At $\alpha = 2$, this branch exchanges iteration stability with the two outer branches.

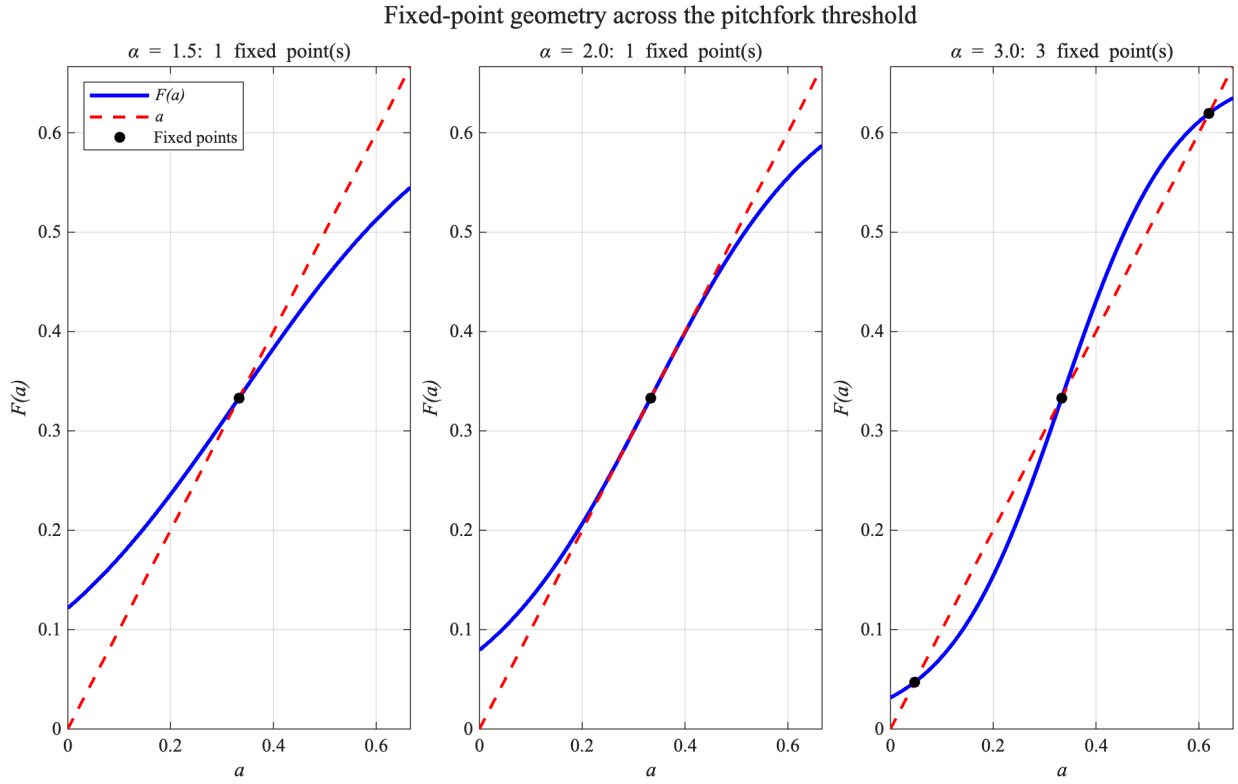


Figure 2: Fixed-point intersections below, at, and above the universal threshold.

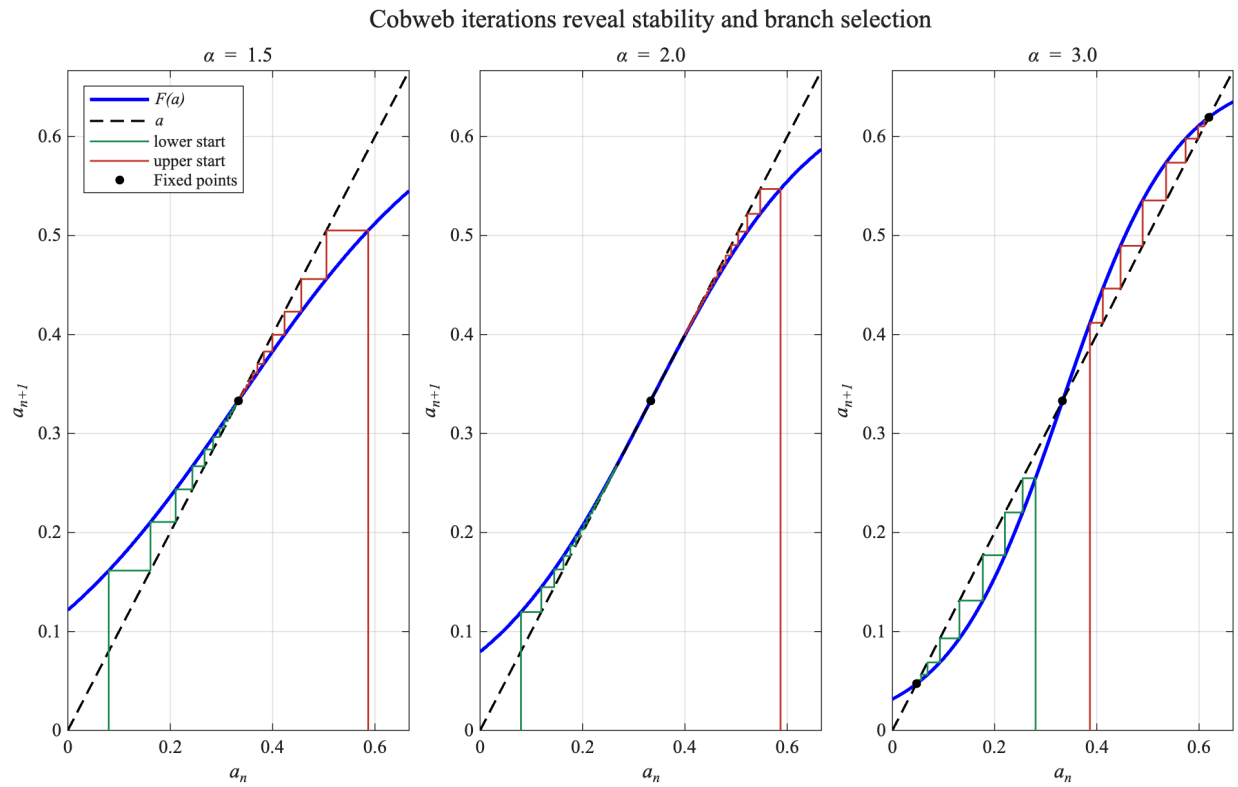


Figure 3: Cobweb paths show convergence, critical slowing, and outer-branch selection. In the $\alpha = 3$ panel, the starts straddle the repelling central fixed point, making the transient motion toward both attracting outer branches visible.

6 Biased law and the S-curve

The bias h in (4) has a direct interface interpretation:

$$\beta_h(s) = e^{2h}\beta_0(s). \quad (17)$$

It rescales the entire Kapitza conductance without changing its dependence on s . In the normalized equation, solve (10) for h :

$$h(x) = \operatorname{artanh}(x) - kx, \quad k = \frac{\alpha}{2}. \quad (18)$$

When $k > 1$, the fold condition is

$$\frac{dh}{dx} = \frac{1}{1-x^2} - k = 0. \quad (19)$$

Define

$$x_f = \sqrt{1 - \frac{1}{k}}, \quad h_c = kx_f - \operatorname{artanh}(x_f) > 0. \quad (20)$$

The two folds are

$$(h, x) = (h_c, -x_f), \quad (h, x) = (-h_c, x_f). \quad (21)$$

Consequently, $|h| < h_c$ gives three distinct roots, $|h| = h_c$ gives two distinct roots with one double root, and $|h| > h_c$ gives one root.

For $\alpha = 3$, $k = 1.5$, and

$$x_f = 0.577350269190, \quad h_c = 0.207546455322. \quad (22)$$

The programs choose $h = h_c/2 = 0.103773227661$, producing the roots in Table 2.

Table 2: Biased roots for $\delta = 0.5$, $\alpha = 3$, and $h = h_c/2$.

x	a	s	$\beta_h(s)$	q	λ	Iteration
-0.798116	0.067295	0.899058	0.037425	0.033647	0.544516	attracting
-0.214294	0.261902	0.607147	0.215683	0.130951	1.431117	repelling
0.894934	0.631645	0.052533	6.011867	0.315822	0.298641	attracting

The same three states can be viewed either as intersections of the biased map with the identity or as intersections of the two heat-flux curves

$$q_K(s) = \beta_h(s)s, \quad q_{\text{bulk}}(s) = \frac{\delta}{1+\delta}(1-s). \quad (23)$$

To retain α as the bifurcation parameter, fix a small nonzero bias h and solve the steady equation for α . For $x \neq 0$, the branch has the exact parameterization

$$\alpha(x; h) = 2 \frac{\operatorname{artanh}(x) - h}{x}. \quad (24)$$

The nonzero bias breaks the reflection symmetry $x \mapsto -x$. Consequently, the perfect pitchfork no longer occurs: the branch favored by the sign of h continues smoothly, while an attracting–repelling pair on the opposite side is created at a saddle node. Combining the steady equation with the tangency condition gives

$$\alpha_{\text{SN}} = \frac{2}{1 - x_{\text{SN}}^2}, \quad h = \operatorname{artanh}(x_{\text{SN}}) - \frac{x_{\text{SN}}}{1 - x_{\text{SN}}^2}. \quad (25)$$

The six continuations below use

$$h \in \{0, 0.002, 0.005, 0.010, 0.020, 0.040\}.$$

Their corresponding multiplicity thresholds are

$$\alpha_{\text{SN}} \in \{2.000000, 2.041774, 2.077216, 2.123116, 2.196796, 2.315799\}.$$

Thus even a very small nonzero bias destroys the exact pitchfork, while the saddle node moves progressively to larger α . The displacement is not linear in h . Near $x = 0$, the two terms in the fold equation satisfy

$$\operatorname{artanh}(x) = x + \frac{x^3}{3} + \frac{x^5}{5} + O(x^7), \quad (26)$$

$$\frac{x}{1-x^2} = x + x^3 + x^5 + O(x^7). \quad (27)$$

Their linear terms cancel, leaving

$$h = -\frac{2}{3}x_{\text{SN}}^3 + O(x_{\text{SN}}^5). \quad (28)$$

For $h \rightarrow 0^+$ the relevant saddle node has $x_{\text{SN}} < 0$, so

$$x_{\text{SN}} \sim -\left(\frac{3h}{2}\right)^{1/3}, \quad \alpha_{\text{SN}} = 2 + 2x_{\text{SN}}^2 + O(x_{\text{SN}}^4). \quad (29)$$

Consequently,

$$\alpha_{\text{SN}} - 2 \sim 2\left(\frac{3h}{2}\right)^{2/3}. \quad (30)$$

This $h^{2/3}$ law is the local cusp scaling associated with the imperfect pitchfork.

The normalized equation is invariant under the simultaneous reflection

$$(x, h) \mapsto (-x, -h). \quad (31)$$

Since $a = \gamma(1 + x)/2$, changing the sign of the bias therefore reflects the entire continuation diagram about $a = \gamma/2$:

$$a(\alpha; -h) = \gamma - a(\alpha; h), \quad \alpha_{\text{SN}}(-h) = \alpha_{\text{SN}}(h). \quad (32)$$

Thus a negative bias favors the lower branch, while the saddle-node pair moves to the upper side. Its threshold depends on $|h|$, not on the sign of h .

In the separate S-curve continuation below, α is fixed and h is the continuation parameter, whereas a remains the steady-state variable. Figure 8 fixes one value of h and displays the resulting map as a function of a ; its cobweb axes therefore use a_n and a_{n+1} .

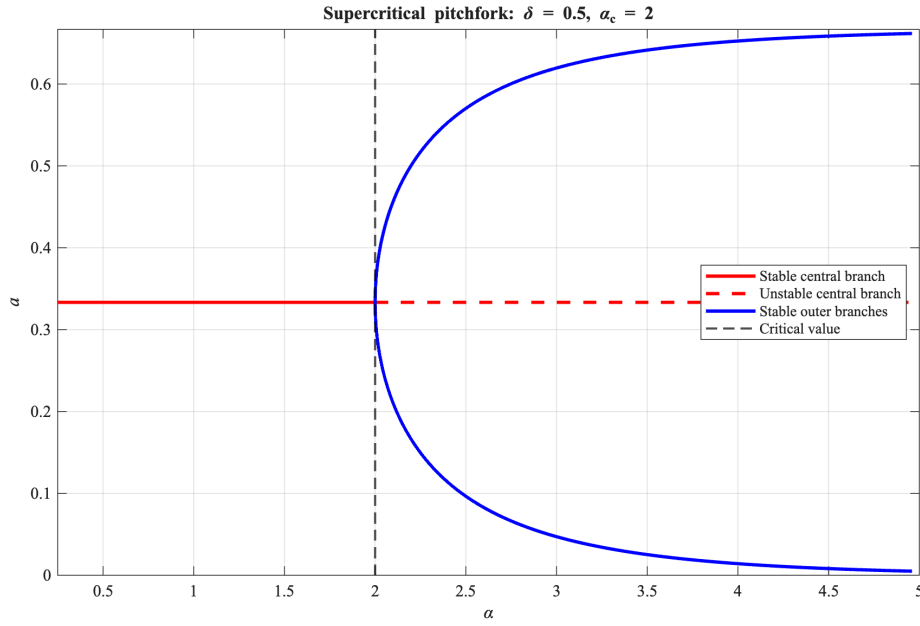


Figure 4: Analytic pitchfork for $\delta = 0.5$. Solid branches are attracting for fixed-point iteration; the dashed central segment is repelling.

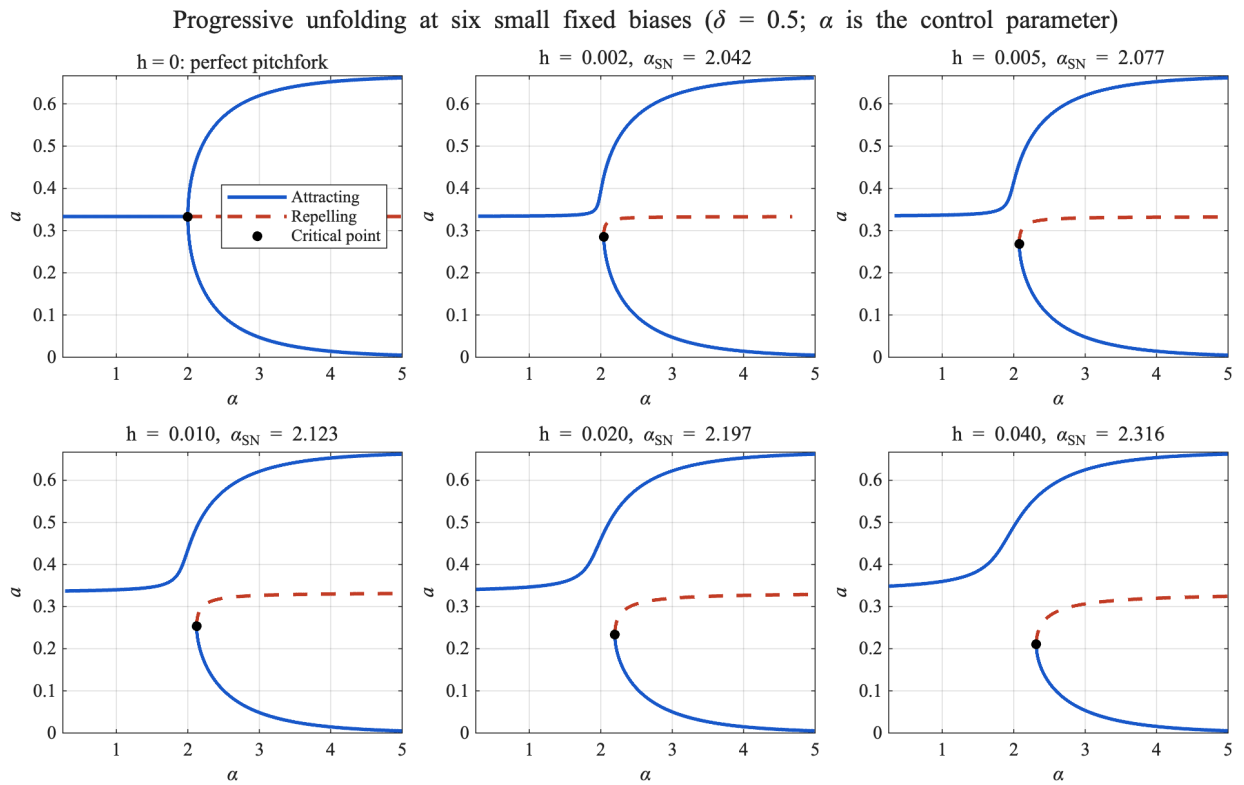


Figure 5: Continuation in α for six small fixed biases. The exact pitchfork exists only at $h = 0$. For every $h > 0$, the favored branch persists smoothly, while the other two branches meet at a saddle node whose location moves to larger α as h increases. Solid and dashed curves denote attracting and repelling fixed points of the scalar iteration, respectively.

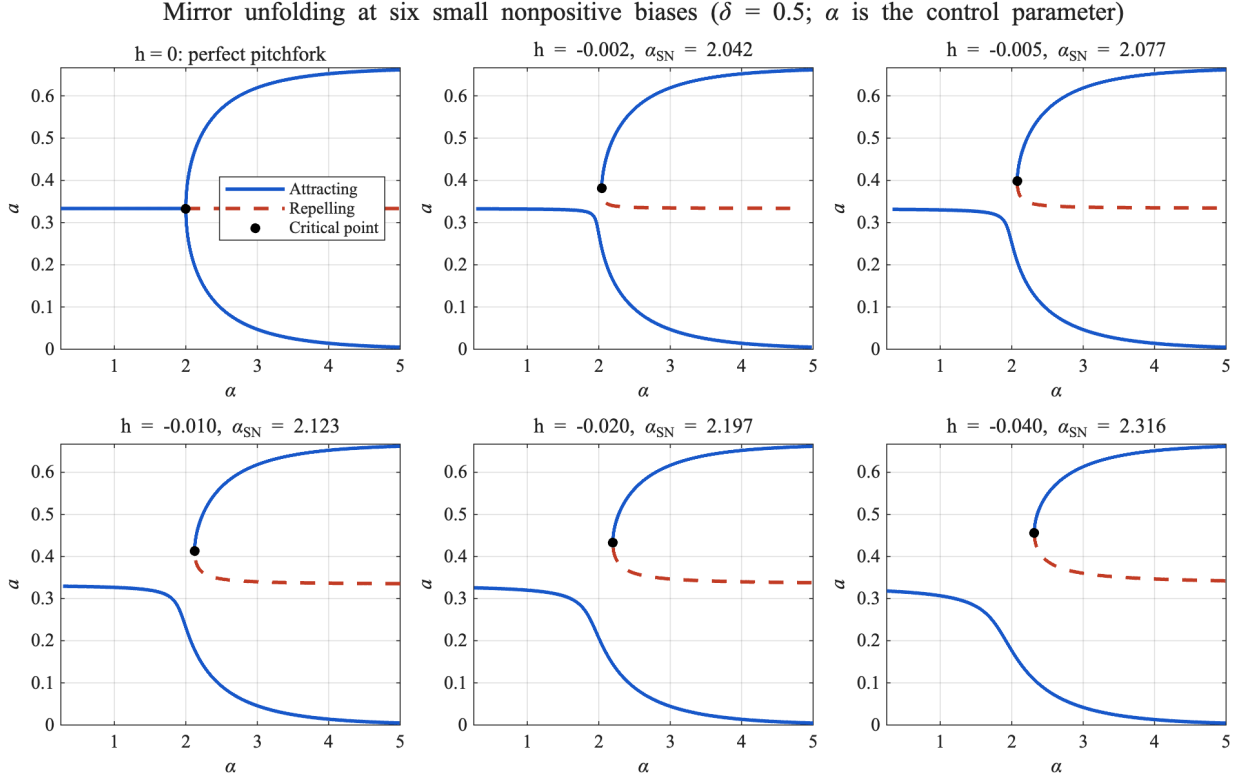


Figure 6: Continuation in α for the reflected sequence $h = 0, -0.002, -0.005, -0.010, -0.020, -0.040$. Negative bias favors the lower attracting branch, and the attracting–repelling pair is created at an upper saddle node. For each matching value of $|h|$, the saddle-node threshold is identical to that in Figure 5, while its state coordinate is reflected through $a = \gamma/2$.

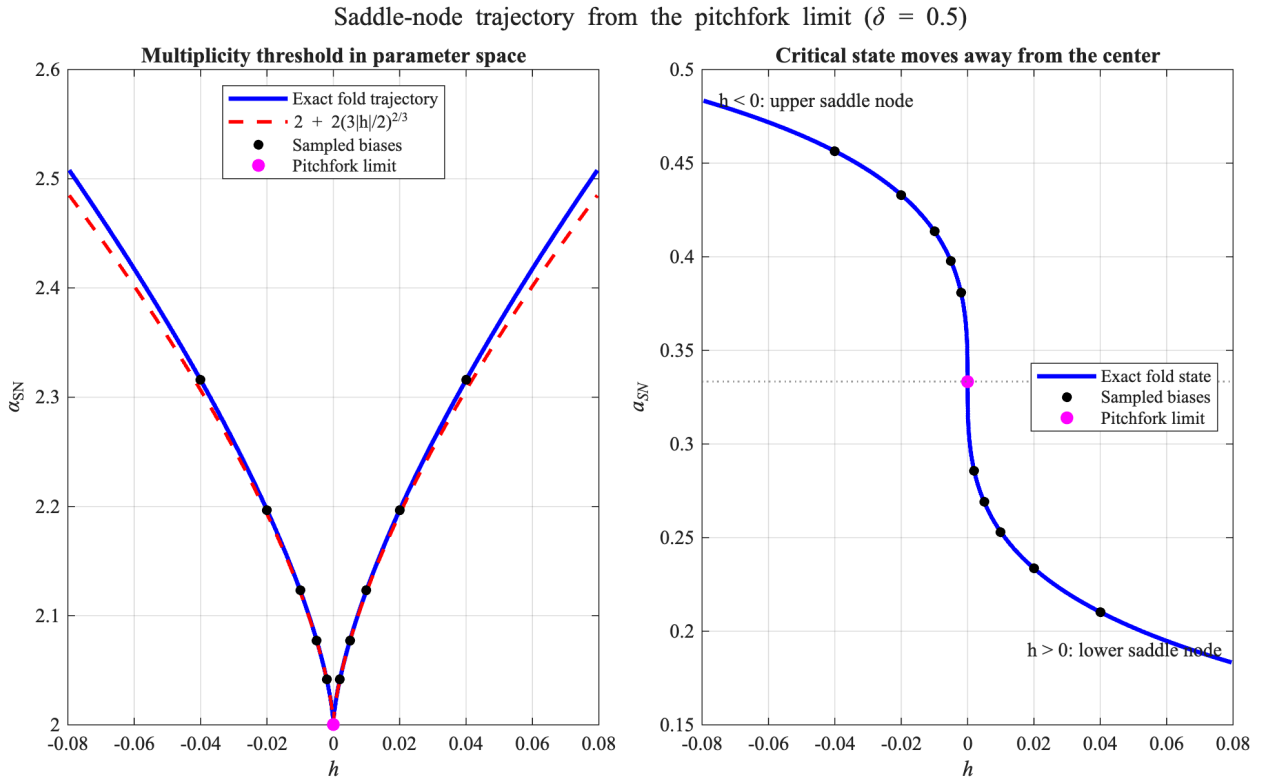


Figure 7: Trajectory of the saddle-node point that replaces the pitchfork critical point. The left panel shows the exact multiplicity boundary in the (h, α) plane together with its local $|h|^{2/3}$ approximation. The right panel shows the corresponding state coordinate: positive and negative biases move the saddle node below and above $a = \gamma/2$, respectively.

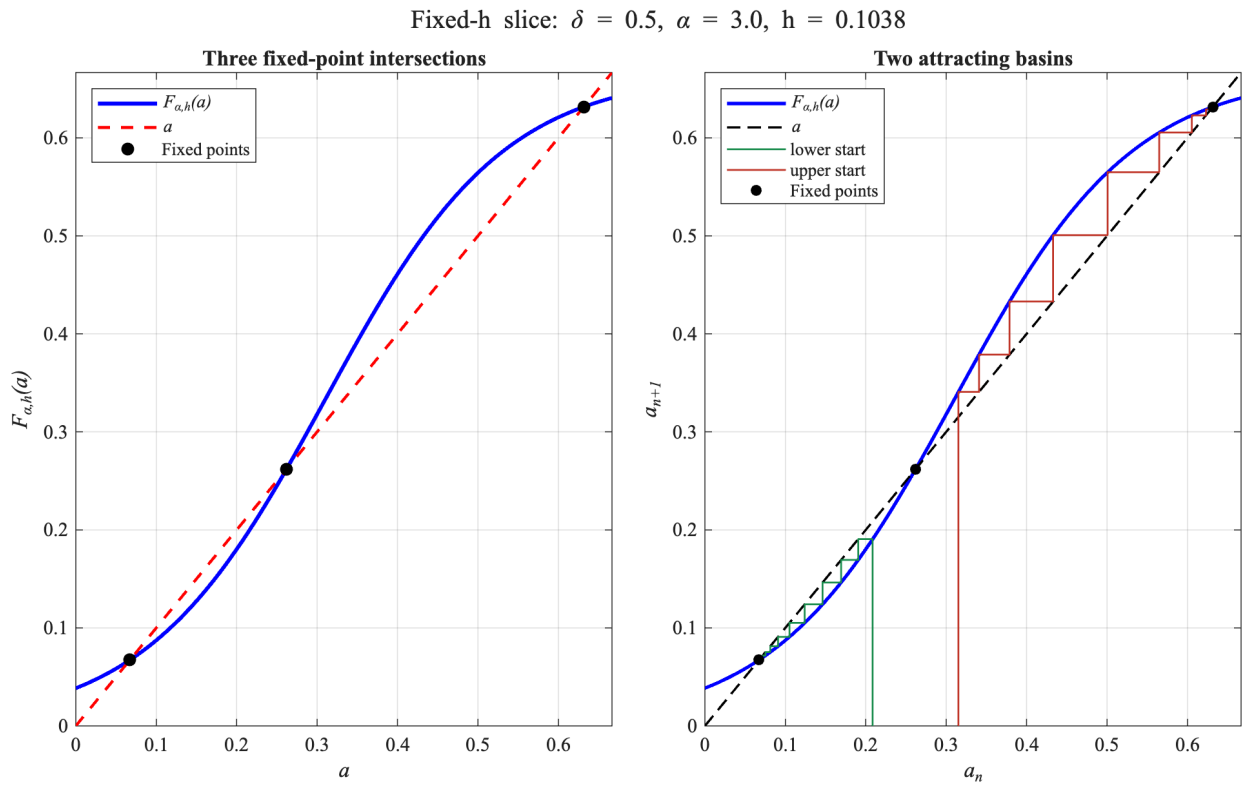


Figure 8: A fixed- h slice of the biased model shown as fixed-point intersections and as two attracting cobweb basins. Here h is the S-curve control parameter and a is the iterated state variable. The starts are placed on opposite sides of the repelling middle fixed point so that both transient paths remain visible.

There is therefore no coordinate error in Figure 8: once h is fixed, the autonomous map is $a_{n+1} = F_{\alpha,h}(a_n)$, so its fixed-point geometry and cobweb are necessarily drawn in the state variable a . A complementary inverse calculation may instead prescribe $a = a_*$ and recover the compatible bias. With $x_* = 2(1 + \delta)a_* - 1$, define

$$R_{a_*}(h) = \tanh(kx_* + h) - x_*. \quad (33)$$

The unique recovered value is

$$h_* = \operatorname{artanh}(x_*) - kx_*. \quad (34)$$

For visualization, the programs use the relaxed parameter iteration

$$h_{n+1} = \mathcal{H}_{a_*}(h_n) = h_n + x_* - \tanh(kx_* + h_n). \quad (35)$$

For the selected upper state, $a_* = 0.631645$ and $h_* = 0.103773$. This is a numerical parameter-recovery algorithm, not a time-dependent physical model; its convergence does not classify physical stability.

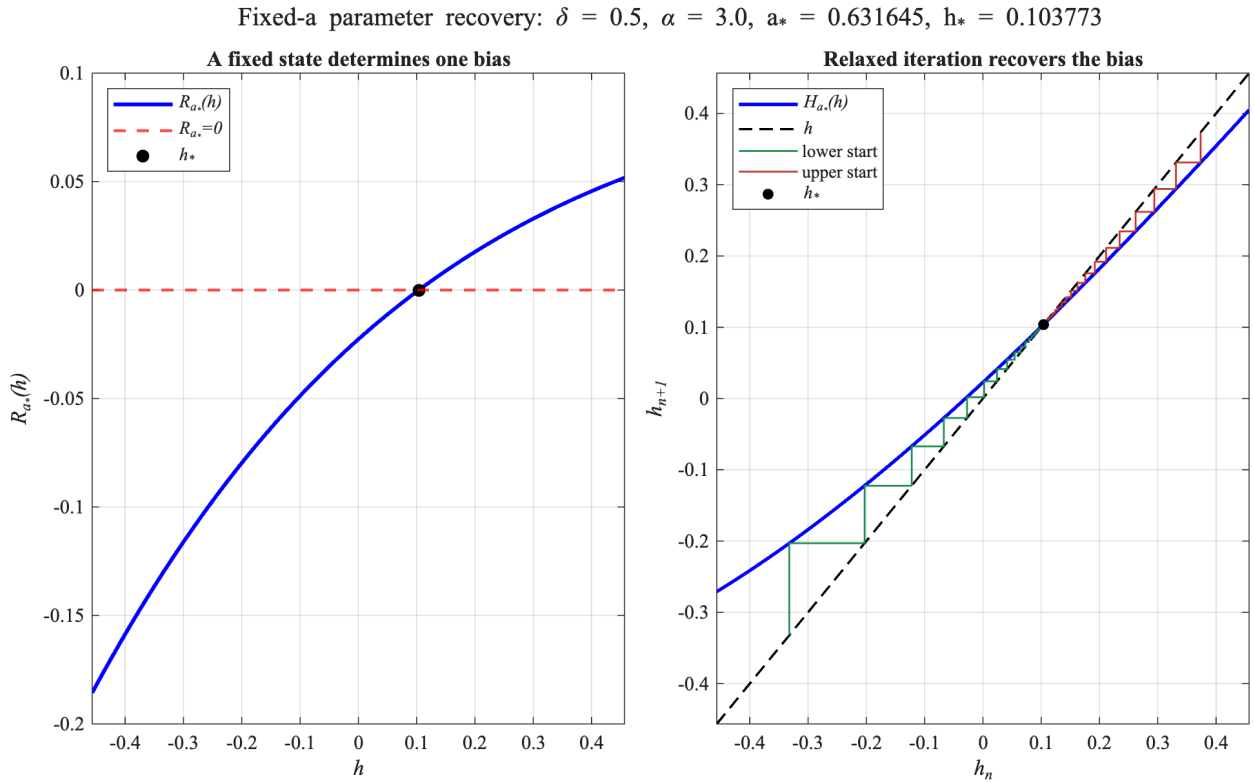


Figure 9: Fixed- a recovery of the compatible bias. The left panel solves the scalar residual, and the right panel visualizes a relaxed iteration in h . This parameter inversion is distinct from both the physical a -iteration and the S-curve bifurcation.

The actual bifurcation diagram with h as continuation parameter is the S-curve in Figure 10. Its horizontal range is restricted to $[-2.2h_c, 2.2h_c]$ so that the two saddle-node folds are visually prominent.

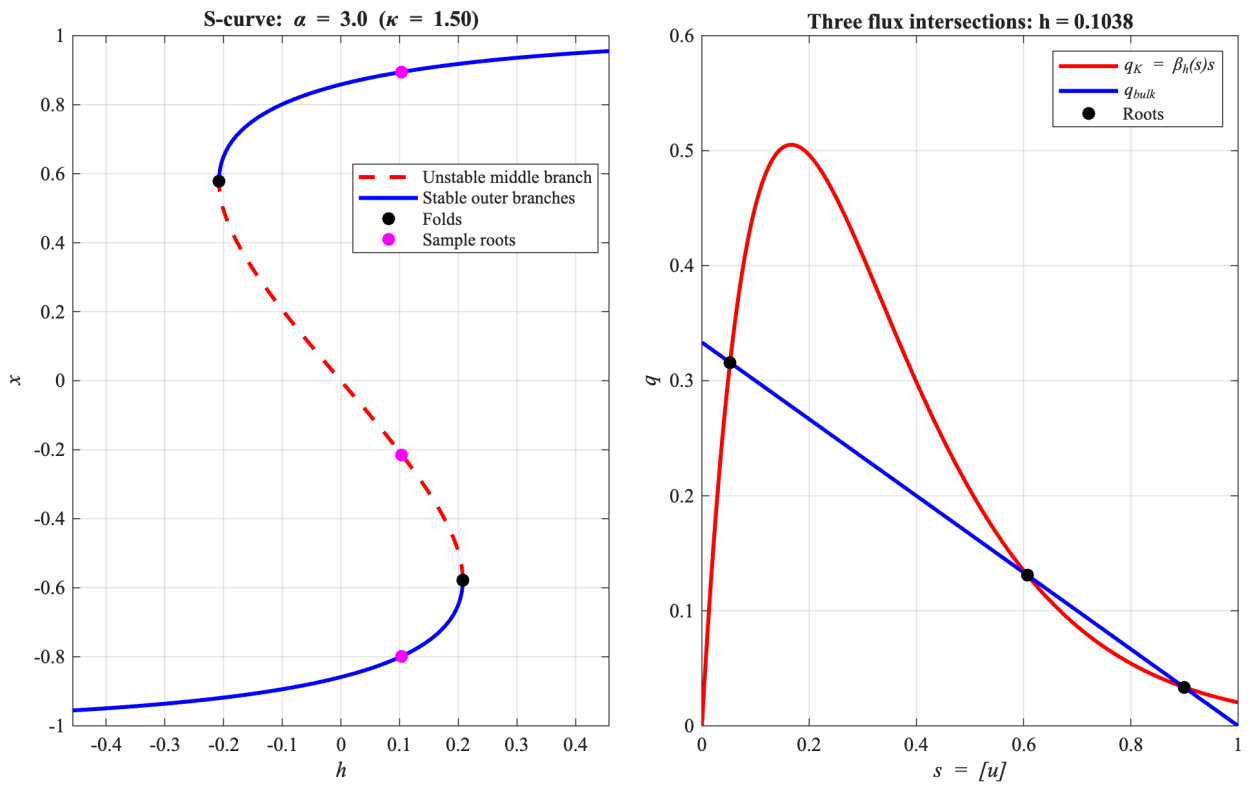


Figure 10: S-curve continuation and the equivalent three heat-flux intersections for the sample bias.

7 How the numerical algorithms match the analysis

Both implementations use the same four computational stages:

1. evaluate $\beta_h(s)$ and the map $F_{\alpha,h}(a)$;
2. solve the normalized residual $R(x) = \tanh(\alpha x/2 + h) - x$;
3. reconstruct a , s , β , q , and λ ;
4. generate intersections, cobwebs, bifurcation curves, S-curves, and data files.

For $\alpha > 2$, the residual has stationary points at $x = \pm\sqrt{1 - 2/\alpha}$. The code inserts those points into the search grid, making each search interval monotone. It then retains roots at interval endpoints, applies a bracketed scalar solver to every sign-changing interval, sorts the candidates, and removes duplicates. This construction detects both ordinary roots and the double root at a fold.

The Python implementation is one command-line module:

```
python3 code/python/kapitza_numerical.py \
  --delta 0.5 --alpha 3 --output-dir output/python
```

It additionally writes CSV files containing all reconstructed states and iteration multipliers. The MATLAB implementation is one script containing local functions:

```
run('code/matlab/kapitza_numerical.m')
```

It reproduces the teacher example and generates the six presentation-quality figures under `output/matlab/`. Both source files use `delta`; comments explicitly state that the teacher's `d` is the same parameter.

8 Conclusions

The implemented model is completely characterized by the exponential conductance (4). It converts the nonlinear transmission problem exactly into the normalized scalar equation (10). In the teacher's alpha convention, the symmetric threshold is universally $\alpha_c = 2$, independent of δ . The central state is unique below threshold; above threshold it becomes repelling for fixed-point iteration and two attracting outer states form a supercritical pitchfork. Multiplying the conductance by e^{2h} breaks symmetry and produces an analytic S-curve with two folds. The Python and MATLAB programs reproduce the same roots, branch classification, cobweb behavior, and heat-flux intersections.

The only stability claim made here is stability of the scalar fixed-point algorithm. Establishing physical stability would require a specified time-dependent transmission model and a separate spectral or energy analysis.